

Q1	A	B	C	D	E	F
Q2	A	B	C	D	E	F
Q3	A	B	C	D	E	F
Q4	A	B	C	D	E	F
Q5	A	B	C	D	E	F
Q6	A	B	C	D	E	F
Q7	A	B	C	D	E	F
Q8	A	B	C	D	E	F
Q9	A	B	C	D	E	F
Q10	A	B	C	D	E	F
Q11	A	B	C	D	E	F
Q12	A	B	C	D	E	F
Q13	A	B	C	D	E	F
Q14	A	B	C	D	E	F
Q15	A	B	C	D	E	F
Q16	A	B	C	D	E	F
Q17	A	B	C	D	E	F
Q18	A	B	C	D	E	F
Q19	A	B	C	D	E	F

1. Which object acts as the power source in a simple battery-operated circuit?

- A. A switch
- B. A clothespin
- C. An alarm buzzer
- D. A light bulb
- E. A battery
- F. A piece of string

2. What do some switches insert between two wires to stop the flow of electricity?

- A. Connecting wires
- B. Insulators
- C. Conductors

D. Power sources

E. Extra batteries

F. Metal components

3. In the analogy of children holding hands in a circle, what do the hands and arms represent in an electrical circuit?

- A. The emergency signal
- B. The power source
- C. Wires connecting the parts
- D. The device needing power
- E. The insulators
- F. The switch

4. Which of the following is a common daily use for an alarm?

- A. Keeping a circuit perfectly separated
- B. Insulating rubber from a power source
- C. Changing the battery in a flashlight
- D. Reminding you to take dinner out of the oven
- E. Heating up toast for breakfast
- F. Turning off a bedroom light

5. What must happen to a circuit for a connected light bulb to turn on?

- A. All parts must be wired together in a continuous, connected loop

B. The power source must be disconnected

C. The wires must be separated by an insulator

D. A buzzer must be added to the wiring

E. A switch must separate the two main wires

F. The battery must be placed outside the loop

6. During the redesign process, what is typically done immediately after deciding what changes to make and drawing the new design?

A. Hiding the alarm materials

B. Purchasing new treasures

C. Revising or re-creating the model

D. Brainstorming initial concepts

E. Showing it to the entire class

F. Evaluating the initial results

7. What is the primary function of a power source in a circuit?

A. To break the loop of electricity

B. To sound a loud buzzer during an emergency

C. To separate two electrical components securely

D. To keep valuable items safe from intruders

E. To act as a protective layer over wires

F. To provide the energy needed for devices to work

8. What is an electrical circuit?

A. A piece of rubber used to block electricity

B. A connected loop that allows power to flow

C. A type of battery used in households

D. A switch that automatically controls lights

E. An unconnected wire dangling from a ceiling

F. A device used strictly for sounding emergencies

9. What can trigger a safety alarm in certain vehicles?

A. Drifting into another lane by accident

B. Keeping valuable items safe

C. Disconnecting the battery

D. Turning on the radio

E. Using the windshield wipers

F. Turning on the bedroom light

10. What term describes materials that do not allow electricity to flow?

A. Buzzers

B. Power sources

C. Conductors

D. Connectors

E. Circuits

F. Insulators

11. When an alarm is triggered by smoke inside a building, what is its main function?

A. To test the wire cutters

B. To release insulators into the air

C. To signal an emergency

D. To interrupt the main power grid

E. To shut off the circuit instantly

F. To act as a power source

12. What is the primary purpose of an alarm?

A. To create new loops in electrical circuits

B. To teach visual literacy in classrooms

C. To keep people or valuable items safe

D. To replace standard light bulbs

E. To provide energy to a battery

F. To block the flow of electricity indefinitely

13. How does a simple door alarm circuit function when the door opens?

A. The buzzer disconnects from the wall

B. The battery drops onto the floor to break the circuit

C. The alarm gets blocked by an insulator

D. Opening the door flips a glass switch

E. The door cuts the wires with a hidden blade

F. A moving wire touches another wire to complete the circuit

14. What is the first step in the formal design process?

A. Triggering the alarm buzzer

B. Evaluating the flaws

C. Building the final model

D. Testing the prototype

E. Describing, drawing the design, and listing materials

F. Revising the broken components

15. Which of the following is considered a good insulator?

A. A metal paperclip

B. A metal brad

C. Copper wire

D. Glass

E. A silver coin

F. Aluminum foil

16. What happens when a circuit is interrupted?

A. The wires become permanently damaged

B. A switch immediately changes the direction of flow

C. The alarm buzzer gets significantly louder

D. The power source doubles its output

E. The circuit automatically redesigns itself

F. The power stops flowing and the device turns off

17. What is the testable goal when creating a protective alarm for a box?

A. The wires break off once the box is lifted

B. The battery recharges when the lid is shut

C. The alarm buzzes when someone opens the box

D. The box automatically locks itself permanently

E. The box changes color when handled

F. The alarm produces a bright flash of light

18. Which component is typically built in by engineers to easily connect or separate two wires?

A. A plastic tube

B. A switch

C. A piece of string

D. A glass pane

E. A metal brad

F. A rubber band

19. What are the standard sequential steps in the basic design process?

A. Plan, Create, Test, Evaluate

B. Test, Plan, Evaluate, Redesign

C. Plan, Evaluate, Create, Test

D. Create, Redesign, Revise, Test

E. Evaluate, Create, Plan, Test

F. Create, Test, Plan, Evaluate